



DONGXING WASTE INCINERATION PLANT LEACHATE BIOGAS HARMLESS TREATMENT AND COMPREHENSIVE UTILIZATION PROJECT



Document Prepared by LGAI Technological Center, S.A. (Applus+ Certification)

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Report Title	Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project
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Summary:

LGAI Technological Center, S.A. (hereafter referred to as "Applus+ Certification") has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of project activity "Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project" (hereafter referred to as "the project activity") and reported in the Project Description.

The project activity is a biogas recovery and power generation project located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China, of which the main purpose is to use biogas for electricity generation. The total installed capacity of the project activity is 3 MW consisting of 3 sets of 1,000 kW generators. The project uses biogas generated by the anaerobic treatment section of leachate for power generation. During the 10 years fixed crediting period, the annually power generated by the project is expected to be 118,470 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the biogas recovered by project, the project activity avoids the emission of methane that would be generated from the anaerobic treatment section of leachate. It's estimated that the project activity could achieve GHG emission reductions of 663,100 tCO₂e during the 10-year fixed crediting period.

The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- AMS-III.H: Methane recovery in wastewater treatment, version 19.0
- AMS-I.D: Grid connected renewable electricity generation, version 18.0
- Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
- Tool to calculate the emission factor for an electricity system, version 07.0
- Demonstration of additionality of small-scale project activities, version 13.1
- Tool for the demonstration and assessment of additionality, version 07.0.0

The projects compliance with, the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline and standard version 4.4

- CDM Validation and Verification Standard for project activities version 03.0
- VCS standard version 4.4
- VCS guideline version 4.3

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified emission reductions (VERs).

A risk-based approach has been followed to perform this validation and verification activity. In the course of Validation, 2 Corrective Action requests (CARs), 0 Clarification Requests (CLs), and 0 Forward action request (FARs) were raised and successfully closed. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and project owners have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfilment of the stated criteria of VCS.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center, S.A. (hereafter referred to as "Applus+ Certification") has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of project activity "Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project" (hereafter referred to as "the project activity") and reported in the Project Description.

LGAI Technological Center, S.A. as the validation body of the project activity has been accredited as a DOE by UNFCCC and also meets the competence requirements as set out in ISO 14065:2020.

The objective of this validation is to ensure that reported information in the Project Description of "Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project" is complete and accurate in accordance with applicable VCS standards and relevant UNFCCC requirements.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the project design (PD). The PD is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board, including the approved baseline and monitoring methodology AMS-III.H-version 19.0 and AMS-I.D version 18.0. The validation was based on the requirements in the CDM Validation and Verification standard for project activities version 03.0 and VCS guideline version 4.3 and standard version 4.4.

The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

1.3 Level of Assurance

The validation report is based on the VCS-PD, supporting evidences made available to the validator and information collected through performing interviews and during the on-site audit assessment.

The validation conclusion is assured a reasonable level of assurance.

1.4 Summary Description of the Project

Project title	Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project
Project Participants	Henan BCCY Environmental Energy Co., Ltd. (Project owner, host country, P. R. China)
Location of the project	Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China Geographic coordinates: longitude of 113.57° E and latitude of 34.37° N
Project start date	21/01/2022, the commissioning start date
Applied Methodology/Version	AMS-III.H, Version 19.0 AMS-I.D, Version 18.0
Scope/Technical Area	Scope 01: energy industries (renewable-/non-renewable sources) and sectoral scope 13: Waste handling and disposal

The project activity is a biogas recovery and power generation project located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China, of which the main purpose is to use biogas for electricity generation. The total installed capacity of the project activity is 3 MW consisting of 3 sets of 1,000 kW generators. The project uses biogas generated by the anaerobic treatment section of leachate for power generation. During the 10 years fixed crediting period, the annually power generated by the project is expected to be 118,470 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of Central China Power Grid (CCPG). Meanwhile by utilization the biogas recovered by project, the project activity avoids the emission of methane that would be generated from the anaerobic treatment section of leachate. It's estimated that the project activity could achieve GHG emission reductions of 663,100 tCO_{2e} during the 10-year fixed crediting period.

2 VALIDATION PROCESS

2.1 Method and Criteria

Validation was conducted using Applus+ Certification's procedures in line with the requirements specified in the VCS Standard version 4.4, CDM M&P, the latest version of the CDM Validation and Verification Standard, and relevant UNFCCC requirements and applying standard auditing techniques. No sampling was utilized during the on-site audit as well as validation for project activity.

Applus+ Certification completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of (if applicable):

- Project Details;
- Application of Methodology;
- Estimated GHG Emission Reduction and Removals;
- Monitoring;
- Safeguards etc.

Applus+ Certification validate that the reported information in the Project Description are complete and accurate in question. This involved a on-site audit and a desk review of the Project Design. This Validation Report describes the findings of this assessment.

The information of the assessment team is included in below of this report.

Assessment team

According to the sectoral scopes / technical area and experiences in the sectoral or national business environment, Applus+ Certification has composed a project assessment team in accordance with the appointment rules in Applus+ Certification. The composition of assessment team has to be approved by the Applus+ Certification ensuring that the required skills are covered by the team. The four qualification levels for team members that are assigned by formal appointment rules as below:

- Leader Auditor (LA)
- Auditor (A)/ Auditor Trainee (AT)
- Technical Reviewer (TR)
- Technical Experts (TE)

Name	Qualification	Coverage of scope	Coverage of Technical Area	Host country experience
Doris Dai	LA/TE	Y (1.1 and 13.1)	Y	Y
Simon Shen	TR	Y (1.1 and 13.1)	Y	Y

Doris Dai (Master's Degree in Environmental Sciences, Bachelor's Degree in Environmental Technology) is an Auditor appointed by Applus+ LGAI for the GHG project assessment and auditing. She has more than 6 years of work experience in CDM/VCS project assessment. Before she joined Applus+ LGAI, she has been working for CTI Certification as senior GHG Auditor for 3.5 years.

Simon Shen (Master Degree in Thermal Energy Engineering, Bachelor Degree in Environmental Engineering) is an Auditor appointed by Applus+ LGAI for the GHG project assessment, auditing and technical review. He has more than 6 years of work experience in CDM/GS4GG/VCS project assessment and review with Applus+, apart from the years of experience working as GHG Auditor and ISO 9001/14001 in TUV SUD before he joined Applus+ for 3.5 years. Mr. Simon Shen has extensive experience also as former Applus+ Shanghai CDM Technical Manager.

2.2 Document Review

The VCS project design version 01 dated 26/09/2022, version 04 dated 25/06/2023 were assessed as part of the validation. Relevant documents were reviewed. A detailed documents reviewed are listed in Appendix 1 of the report.

2.3 Interviews

The key personnel interviewed through site visit are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Jinlong Tao	General Manager	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.	Status of the project (including PPs); Applicability of selected methodology;
Weiguang Chang	Operation Director	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.	Baseline of the project and its updates;

Yubo Chen	Staff	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.	Emission factors and their updates; Monitoring plan;
Fei Liang	Villager	Zhengan County	Stakeholder consultation process and its outcomes. The process and participation of the stakeholder consultation;
Ling Xu	Villager	Zhengan County	The impact of the project activity; The complaint by local stakeholders and the implementation of the mitigation measures.
Lei Wang	Project Manager	Henan BCCY Environmental Energy Co., Ltd.	Data collection and ER calculation.
Yaya Xu	Project Manager	Henan BCCY Environmental Energy Co., Ltd.	

2.4 Site Inspections

The assessment team performed the on-site validation (Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China) on 26-27/12/2022. The interviewed personnel and objective are listed in above table of section 2.3.

2.5 Resolution of Findings

As an outcome of the validation process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

- Non-compliance with the project design or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;
- Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the project participants;
- Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM or VCS requirements have been met.

All CARs and CLs raised during validation shall be resolved prior to submitting a request for issuance.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design and Monitoring report. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation and verification process, the concerns raised, and responses given are summarized below in the appendix 2.

The final VCS PD Version 04 submitted by project developer on 25/06/2023 serves as the basis for the final assessment presented. Additional changes to the project during the validation and verification process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
Project design document and Monitoring report	00	00	00
Description of project activity	00	00	00
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	00	00	00
- Deviation from methodology	00	00	00
- Clarification on applicability of methodology, tool and/or standardized baseline	00	00	00
- Demonstration of additionality	00	00	00
- Emission reductions	00	01	00
- Monitoring plan	00	01	00
-Stakeholders consultation process	00	00	00
- Public comments	00	00	00
Others (please specify)-Matter related to double counting- for validation	00	00	00
Total	00	02	00

The list of findings and their resolution is presented in appendix II of this report.

2.5.1 Forward Action Requests

None FAR was raised during the validation process.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project

The project activity is a biogas recovery and power generation project of which the main purpose is to use biogas for electricity generation. The total installed capacity of the project activity is 3 MW consisting of 3 sets of 1,000 kW generators.

The project uses biogas generated from the anaerobic treatment section of leachate for power generation. It was difficult to collect and treat biogas, the biogas generated by the UASB was emitted to the atmosphere directly without any recovery and utilization. Before the implementation of the project, Zhengzhou (East) environmental Protection Energy Project invested by Zhengzhou Dongxing Environmental Protection Energy Co., Ltd. incinerates domestic waste with a daily treatment capacity of 4,000 t/d. Leachate will be produced during the storage and fermentation process in the garbage bin. The supporting treatment system of the factory is "pretreatment +UASB (upflow anaerobic sludge blanket) + secondary A/O-MBR aerobic system + nanofiltration NF+STRO" treatment process. A large amount of biogas is produced in the anaerobic treatment section of UASB. The Zhengzhou (East) environmental Protection Energy Project started to construct in 2018 and started operation in 07/2019, because it was difficult to collect and treat biogas, the biogas generated by the UASB was emitted to the atmosphere directly without any recovery and utilization. The biogas was being vented to the atmosphere in 07/2019 when Zhengzhou (East) environmental Protection Energy Project started operation. During the 10 years fixed crediting period, the power generated by the project is expected to be 118,470 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of CCPG. Meanwhile by utilization the biogas recovered by project, the project activity avoids the emission of methane that would be generated under anaerobic treatment section of leachate. It's estimated that the project activity could achieve GHG emission reductions of 663,100 tCO₂e during this crediting period.

The project activity is located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China. The project activity has been developed by Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd. The geographic coordinates of the project activity is longitude of 113.57° E and latitude of 34.37° N which is confirmed by on-site visit.

The Project activity has been approved by China's government by checking Project Approval and Environmental Impact Assessment (EIA) approval. The project started construction in 15/09/2021 confirmed by construction order and commissioned on 21/01/2022 by checking operation log and on-site visit interview.

Project design, including eligibility criteria for grouped projects

The technical specifications are mentioned as below and the same were checked during on-site visit:

Main Equipment	Parameter	Value
Gas engine	Type	J320GS
	Manufacturer	INNIO Jenbacher GmbH & Co OG
	Rated capacity	1,000 kW
	Rated rotation speed	1,500 r/min
	Lifetime	20 year

This is not a grouped project activity. Thus, this section is not applicable for this project.

By checking Project approval, EIA approval and on-site visit, assessment team confirmed the project has been designed to include a single installation of an activity.

Project proponent and other entities involved in the project

Assessment team checked through on-site audit and confirms that the details of the project proponent is as below:

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Project Manager
Address	8/F, Hengmei Business Building, 22 Dongfeng Road, Zhengzhou, China
Telephone	+86 371 56737901
Email	lwang@bccynewpower.com

Assessment team checked through on-site audit and confirms that the details of the other entity involved is as below:

Organization name	Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd.
Role in the project	Project Operation Entity
Contact person	Yaya Xu

Title	Project Manager
Address	8/F, Hengmei Business Building, 22 Dongfeng Road, Zhengzhou, China
Telephone	+86 371 56737901
Email	xuyaya@bccynewpower.com

Project Ownership

Henan BCCY Environmental Energy Co., Ltd. Zhongmu Branch is the project owner of project activity, and its mother company Henan BCCY Environmental Energy Co., Ltd. is project proponent (PP) of project activity and they have the legal right to control and operate the project activities

The project ownership has been checked by the Assessment Team and demonstrated through checking business license and project approval.

Project Start Date

Start date of the project activity is 21/01/2022, which is the commissioning start date. Assessment team checked operation log to confirm the date.

Project crediting period Date

The project activity adopts fixed crediting period of 10 years period.

Assessment team confirms that the crediting period dates for the project is as below:

Crediting Period Start date: 21/01/2022

Crediting Period End date: 20/01/2032

Project Scale and Estimated GHG Emission Reductions or Removals

Assessment team confirms that the project is a small-scale project under CDM scheme. The total installed capacity of the project activity is 3 MW.

Project Scale	
Project	✓
Large project	

As the estimated annual average GHG emission reductions or removal per year is 66,310 tCO₂e which is less than 300,000 tonnes of CO₂e per year, thus the project falls in the category of Project.

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
21/01/2022-31/12/2022	62,677
2023	66,310
2024	66,310
2025	66,310
2026	66,310
2027	66,310
2028	66,310
2029	66,310
2030	66,310
2031	66,310
01/01/2032-20/01/2032	3,633
Total estimated ERs	663,100
Total number of crediting years	10
Average annual ERs	66,310

By checking cooperation agreement between owner of Dongxing Waste Incineration Plant and Henan BCCY Environmental Energy Co., Ltd. which is the mother company of Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd. regarding to the utilization of biogas, the cooperation period for biogas generation will last after 20/01/2032.

The above estimated emission reduction is confirmed by assessment team via emission reduction calculation spreadsheet. The calculation is conservative and this acceptable to the assessment team.

Project location

The project activity is located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P. R. China. The project activity has been developed by Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd. The project is located at longitude of 113.57° E and latitude of 34.37° N which is confirmed by checking on-site visit.

Conditions prior to project initiation

Before the implementation of the project activity, the electricity generated by the project would be supplied by CCPG in the baseline scenario and the biogas combusted would be disposed without utilization.

Project compliance with applicable laws, statutes and other regulatory frameworks

Assessment team confirms that the Project has been approved by P. R. China's government by checking the Project approval and Environmental Impact Assessment (EIA) approval.

By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

Participation under other GHG programs

Projects registered (or seeking registration) under other GHG program(s)

The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. Applus+ Certification checked the REC Mechanism database of P. R. China and found that the project activity is not accredited / registered under REC mechanism. Further, declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:

- <http://www.irecstandard.org/>
- <http://cdm.unfccc.int/>
- <http://www.goldstandard.org/>

Rejection by other GHG programs

The Project is not rejected by other GHG programs. A declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:

- <http://www.irecstandard.org/>
- <http://cdm.unfccc.int/>
- <http://www.goldstandard.org/>
- <http://verra.org/>

Other forms of credit

Emissions trading programs and other binding limits

Applus+ Certification confirms that the Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. Applus+

Certification checked International REC Mechanism and found that the project activity is not accredited/ registered under REC mechanism. Further, Declaration in effect of the same has been submitted by project proponent to audit team and found to be correct. Thus, it is concluded that the project activity not involved on other Emissions trading programs and other binding limits.

Other forms of environmental credit sought or received and eligible to be sought or received

The Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

Renewable energy certificates are available for trading in the host country. However, the same is not availed by the project proponent. The undertaking regarding the same is submitted by PP which is acceptable to the assessment team and assessment team also checked the REC web site (<http://www.irecstandard.org/>) and found the declaration to be correct.

Additional information relevant to the project

Leakage management for AFOLU projects

Not applicable to the project activity.

Commercially sensitive information

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation and verification.

Sustainable development contributions

The project activity would contribute sustainable development in the region in following aspects confirmed by on-site audit:

Reduce CH₄ emission. The project will recover and destroy leachate biogas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030;

Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in Central China Power Grid and further help to achieve the national action to promote renewable energy development;

Improve the operational safety of the waste incineration plant. The project reduces the potential safety hazards of leachate biogas explosion by recycling leachate biogas and improves the safety of waste incineration plant operation;

The project will also provide new jobs opportunities and increase tax revenue, which will have a positive effect on the local economy.

Overall, it is confirmed that the PD is accurate, complete, and provides an understanding of the nature of the project.

3.2 Safeguards

3.2.1 No Net Harm

The Environmental Impact Assessment (EIA) Report of the project activity compiled by Henan Langtian Environmental Technology Co., Ltd. in September 2021 was approved by Ecological environment Bureau of Zhengzhou County on 15/10/2021. The assessment team confirm all environmental impacts has been analyzed and no net harm was detected.

As all potential environmental and socio-economic impacts have been discussed in the EIA. The assessment team check the same and find there are no negative potential environmental and socio-economic impacts. Moreover, based on the professional experience of assessment team, and on-site audit, the assessment team confirm that there are no negative potential environmental and socio-economic impacts detected during the on-site audit.

3.2.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. The assessment team checked the relevance of the dates during the validation site visit.

A local stakeholder consultation meeting was held on 25/11/2021 by the project owner. Before the stakeholder consultation meeting, the project information with contact information and the stakeholder meeting notice were published on the bulletins at and nearby the project site on 10/11/2021 to invite Local stakeholders.

During the meeting, the project information and the continuous input / grievance mechanism have been introduced to the local stakeholders. The project information with contact information has also been posted on the bulletins at and nearby the project site.

During the meeting, a survey was carried out on the local residents and comments received from the survey are summarized as follows. The survey was conducted through distributing and collecting responses to a questionnaire. In total 30 out of 30 questionnaires verified by checking all questionnaires were returned with a 100% response rate.

By checking the questionnaires, Applus+ Certification confirm that the local stakeholder has no major negative comments for the construction of the project activity, and minor negative comments have been considered with mitigation measurements.

Expect for introducing the project information, grievance mechanism was also explained during the local stakeholder consultation meeting. A grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project Participant will be checking the comments in the book on a regular basis, and record responses. The project

information with contact information has also been posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Participant will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible. The same has been confirmed by site visit interview.

Thus, the assessment team is of the opinion that the stakeholder meeting was adequate and appropriate.

3.2.3 Environmental Impact

By checking Environmental Impact Assessment (EIA) Report, the positive environmental impact has been presented as below:

Waste water

The wastewater produced by the project consists of biogas condensation waste water, circulating cooling waste water of generator set, and reclaimed waste water of soft water equipment. Condensation wastewater will be treated by leachate treatment system of Zhengzhou Dongxing Environmental Protection Energy Co., LTD., the sewage is reused and not discharged. Therefore, it can meet the standards of "The reuse of urban recycling water- Waste quality standard for industrial uses"(GB/T19923-2005).

Waste Gas

The exhaust gas produced by this project is mainly the exhaust gas of the generators. The concentration of the exhaust gas (mainly SO₂, NO_x and particulate matter) after combustion in gas-generator sets and processed by the SCR denitration system is discharged through the 15 m high cylinder exhaust.

Noise pollution

According to the prediction results, the acoustic environment quality of the project equipment noise transmitted to the northwest boundary of project can meet the Class 2 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008). The acoustic environment quality of the project equipment noise transmitted to the southeast boundary of project can meet the Class 4 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008).

Solid waste

The solid waste of the project is mainly sulfur paste, waste oil, waste oil barrel and waste denitration catalyst. The sulfur paste is stored in the sulfur storage room and sold regularly. The waste oil, waste oil barrels and waste denitration catalyst are stored in the hazardous waste room of the factory, and regularly handed over to qualified units for recycling and processing. The

construction standard of hazardous waste room shall meet the requirements of "Standard for pollution control on hazardous waste storage" (GB18597-2001).

3.2.4 Public Comments

Applus+ Certification noted that this project was open for public comment from 29/11/2022 to 29/12/2022. The detail was checked by the assessment team in the following web platform: <https://registry.verra.org/app/projectDetail/VCS/3851>.

During the period, no public comments were received.

3.2.5 AFOLU-Specific Safeguards

The project activity is not an AFOLU project. For non-AFOLU projects, this section is not required.

3.3 Application of Methodology

3.3.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below:

Title: Methane recovery in wastewater treatment

Reference: The project activity meets the eligibility criteria of small-scale project as the estimated annual average GHG emission reductions or removal per year is 59,312 tCO₂e under Type III and 6,777 tCO₂e under Type I which are all less than 60,000 tonnes of CO₂e per year.

Methodology: Methane recovery in wastewater treatment¹, AMS-III.H, Version 19.0

Grid connected renewable electricity generation², AMS-I.D, Version 18.0

Type I: Energy industries (renewable / non-renewable sources)

Sectoral scope(s): 01 and 13

Category: Approved Methodology

Tools referred with above methodology and applicable for project activity are:

- Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 03.0
- Tool to calculate the emission factor for an electricity system, Version 07.0

¹ <https://cdm.unfccc.int/methodologies/DB/JPYB4DYQUXQPZLBDVPHA87479EMY9M>

² <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQOOFQOH4SBK>

- Positive lists of technologies, Version 04.0

3.3.2 Applicability

The applicability of AMS-III.H is justified as below:

Applicability condition	Compliance of the project	Assessment
This methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; c) Introduction of biogas recovery and combustion to a sludge treatment system; d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment, to an untreated wastewater stream; f) Introduction of a sequential stage of wastewater treatment with	Applicable. This project collects the biogas generated by the anaerobic treatment section of leachate to generate electricity and connect to the grid. The simplified baseline methodology is applicable to this project activity because without the project activity, methane from the existing anaerobic treatment system would continue to be emitted into the atmosphere. As such, paragraph 2(f) of the methodology applies "Introduction of a sequential stage of wastewater	By checking the FSR, Project approval and on-site audit, it is able to confirm the project collects the biogas generated by the anaerobic treatment section of leachate to generate electricity and connect to the grid. Hence, the project meets the condition of f).

biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery)”	
In cases where baseline system is anaerobic lagoon the methodology is applicable if: (a) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken;	Applicable. a) The baseline system is anaerobic lagoons with the minimum depth of 2 m; b) The average atmospheric temperature in the region is above 15°C from April to October;	By checking FSR, EIA and on-site audit, it is confirmed that the depth of anaerobic lagoons are greater than two meters. By checking public on-line information³, it is confirmed that the annual average temperature is above 15°C from April to October in the project site. By on-site audit, it is confirmed that the sludge is incinerated in the incineration system of this project. There was no sludge

³ https://www.tianqi.com/qiwen/city_zhongmou/

(b) Ambient temperature above 15°C, at least during part of the year, on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	c) The sludge is incinerated in the incineration system of this project. There was no sludge generation in the baseline scenario. Therefore, the project activity is applicable for this criterion.	generation in the baseline scenario.
The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or € Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or	Applicable. The recovered biogas will be utilized to generate electrical energy directly. As such, paragraph 4 (a) applies.	By checking the FSR, Project approval and on-site audit, it is confirmed that the recovered biogas will be combusted for power generation.

(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users; (d) Hydrogen production; € Use as fuel in transportation applications after upgrading		
If the recovered biogas is used for project activities covered under paragraph 4(a), that component of the project activity can use a corresponding methodology under Type I.	Applicable. Since the biogas captured will be used for electrical energy generation directly for this project activity, methodology under Type I should be used for the electricity generation components. Therefore, AMS.I.D is applied for the project activity.	By checking the FSR, Project approval and on-site audit, it is confirmed that the recovered biogas will be combusted for power generation. And utilization of biogas for electricity generation is under Type I.
For project activities covered under paragraph 4(b), if bottles with upgraded biogas are sold outside the project boundary, the end-use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period CO2 emissions avoided by the	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.

displacement of fossil fuel can be claimed under the corresponding Type I methodology, e.g. “AMS-I.C.: Thermal energy production with or without electricity”		
For project activities covered under paragraph 4€(i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.
For project activities covered under paragraph 4€(ii), emission reductions for the displacement of the use of fuels can be claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.
In particular, for the case of paragraph 4(b) and (c)(iii), the physical leakage during storage and transportation of upgraded biogas, as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of “AMS-III.H.: Methane recovery in wastewater treatment” shall be followed in this regard.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.
For project activities covered under paragraph 4(b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.

exist) or, in the absence of national regulations, a minimum of 96 per cent (by volume).		
If the recovered is utilized for the production of hydrogen (project activities covered under paragraph 3(d)), that component of the project activity shall use the corresponding methodology “AMS-III.O.: Hydrogen production using methane extracted from biogas”.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.
If the recovered biogas is used for project activities covered under paragraph 4€, that component of the project activity shall use corresponding methodology “AMS-III.AQ.: Introduction of Bio-CNG in transportation applications”.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.
New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the relevant requirements in the “General guidelines for SSC CDM methodologies”. In addition, the requirements for demonstrating the remaining lifetime of the equipment replaced, as described in the general guidelines shall be followed.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is confirmed that this is not applicable for the project.
The location of the wastewater treatment plant as well as the source generating the wastewater shall be	Applicable. The location of the wastewater treatment plants is	By checking FSR and on-site audit, it is confirmed that the location of the wastewater treatment plant as well as the

uniquely defined and described in the PDD	uniquely defined in the factory site of Dongxing Waste Incineration Plant. The source generating the wastewater is from production water and domestic sewage from waste incineration plant, and the wastewater is being imported to the leachate treatment station treatment through channels.	source generating the wastewater shall be uniquely defined and described in the PD consistent with requirement in the methodology.
Measures are limited to those that result in aggregate emissions reductions of less than or equal to 60 ktCO ₂ equivalent annually from all Type III components of the project activity	The estimated average annual emissions reductions are below 60 ktCO ₂ e	The estimated annual ERs are less than the 60,000 tCO ₂ limit for Type III components.

The applicability of AMS-I.D is justified as below:

Applicability condition	Compliance of the project	Assessment
This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant (s); (c) Involve a retrofit of (an) existing plant (s); (d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or;	Applicable. The project installs a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of	By checking the FSR, Project approval and on-site audit, it is able to confirm the project involved in an installation of a greenfield plant which meet the condition of (a).

(e) Involve a replacement of (an) existing plant(s).	the project, which corresponds to point (a).	
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m²; The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m².	Not applicable. The project is not a hydro power plant.	By checking the FSR, Project approval and on-site audit, it is able to confirm the project is not a hydro power project. Hence not applicable
If the new unit has both renewable and non-renewable components (e.g. a wind/ diesel unit), the eligibility limit of 15 MW for a small-scale VCS project activity applies only to the renewable component. If the new unit co-fires fossil fuel. The capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable. The project does not use non-renewable components nor co-fires fossil fuels. Anyway, the total installed capacity is below 15 MW.	By checking the FSR, Project approval and on-site audit, it is able to confirm the installed capacity of the project is below 15 MW.
Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is able to confirm the project only involved in electricity generation.

	The project only involves electricity generation.	
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable. The project does not involve addition of renewable energy generation units at an existing renewable power generation facility.	By checking the FSR, Project approval and on-site audit, it is able to confirm the project does not involve in addition of renewable energy generation units at an existing renewable power generation facility.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.	Not applicable.	By checking the FSR, Project approval and on-site audit, it is able to confirm the project does not involve in retrofit, rehabilitation or replacement.
In the case of leachate biogas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-L.C. Thermal energy production with or without electricity shall be explored.	Applicable. The project installed 3 engines to combust leachate biogas of Dongxing waste incineration plant, which mainly contains 65% methane.	By checking the FSR, Project approval and on-site audit, it is confirmed that the project installed 3 engines to combust leachate biogas of Dongxing waste incineration plant, which mainly contains 65% methane.
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool	Not applicable.	By checking the FSR, Project approval and on-site audit, it is

"Project emissions from cultivation of biomass "shall apply.		able to confirm the project does not involve in biomass utilization.
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Applicability conditions of "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation"

If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption:

- (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer;
- (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or
- (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.

Emissions are calculated for electricity consumption, Scenario A "Electricity consumption from the grid" applies to the sources of electricity consumption for the project activity.

This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated:

- (a) Scenario I: Electricity is supplied to the grid;
- (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or
- (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.

This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, Scenario I: "Electricity is supplied to the grid" applies to the recipient of the electricity generated.

This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO₂ emissions.

No captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.

Applicability conditions of “Tool to calculate the emission factor for an electricity system”

This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).

The electricity supplied by the project was exported to CCPG. OM, BM and CM are estimated using the tool for calculating baseline emissions for the project activity.

Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option lia and option lib. If option lia is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.

The electricity generated by the project activity was exported to CCPG. The emission factor for the project electricity system is calculated for grid power plants.

In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.

The project activity is located in P. R. China, not in any Annex I country.

3.3.3 Project Boundary

As per AMS-III.H, the project boundary is the physical, geographical site where the leachate biogas is captured and used, in the baseline and project situations. And also includes all the power sources connected to the CCPG, which expands throughout Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province and Chongqing City.

It is confirmed that the implementation of the project would not affected the treatment systems, i.e. sections operating in the project scenario under the same operational conditions as in the baseline scenario. The water treatment system of the factory is "pretreatment +UASB+ secondary A/O-MBR aerobic system + nanofiltration NF+STRO" treatment process. The project adds biogas collection, transmission and treatment system, with subsequent electricity generation and grid

connection system. The wastewater inflow and COD content, temperature, retention time, etc. are not affected by the biogas collection.

It is confirmed by the assessment team that the project boundary of the project activity includes geographical site where the leachate biogas is captured and used. The electricity used by the project activity is sourced from CCPG and electricity generated by biogas utilization of the project activity is connected to CCPG. Therefore, the project boundary includes the whole wastewater and leachate treatment and all grid-connected power plants in CCPG.

The sources and GHG gases involved for the Project activity are as below.

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from wastewater treatment system	CH ₄	Yes	The major source of emissions in the baseline.
		N ₂ O	No	Excluded for simplification.
		CO ₂	No	CO ₂ emissions from the decomposition of organic waste in the wastewater are not accounted for.
	Emission from electricity generation	CO ₂	Yes	The major source of emissions in the baseline.
		CH ₄	No	Excluded for simplification, this is conservative.
		N ₂ O	No	Excluded for simplification, this is conservative.
	Emissions from heat generation	CO ₂	No	No heat generation is included in the project.
		CH ₄	No	No thermal energy generation is included in the project.
		N ₂ O	No	No thermal energy generation is included in the project.
	Emissions from the use of natural gas	CO ₂	No	Excluded for simplification. This is conservative
		CH ₄	No	No supply of leachate biogas is included in the project.
		N ₂ O	No	Excluded for simplification. This is conservative
Project activity	Emissions from fossil fuel consumption for purposes	CO ₂	No	Excluded because there is no fossil fuel consumption.

Source	Gas	Included?	Justification/Explanation
other than electricity generation or transportation due to the project activity	CH ₄	No	Excluded because there is no fossil fuel consumption.
	N ₂ O	No	Excluded because there is no fossil fuel consumption.
	CO ₂	Yes	Main emission source.
	CH ₄	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.
	N ₂ O	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.
	CO ₂	No	Excluded because there is no flaring in the project.
	CH ₄	No	Excluded because there is no flaring in the project
	N ₂ O	No	Excluded because there is no flaring in the project
	CO ₂	No	CO ₂ emissions from the decomposition of organic waste in the wastewater are not accounted for.
	CH ₄	Yes	CH ₄ is the major component in the biogas produced during anaerobic wastewater treatment.
	N ₂ O	No	Excluded for simplification.

3.3.4 Baseline Scenario

As per AMS-III.H, the baseline scenario involves a baseline system with anaerobic wastewater treatment system without biogas recovery,

By on-site audit and checking Project approval, it is confirmed that in the absence of the project, Leachate biogas of Dongxing Waste Incineration Plant is left to decay within the project boundary, and methane is emitted to the atmosphere directly without any recovery and utilization. Also the leachate was treated in the wastewater treatment system. It is confirmed that current treatment is compliant with the legislation in P. R. China, which do not stipulate any type of wastewater technology to be used, except a discharge control of the mill effluent. The baseline scenario is identified as the continuation of current situation, resulting in free release of methane into the atmosphere.

The project activity ultimate biogas to produce electricity and supplies to the grid. The assessment team confirm the baseline of the project is in the absence of the project, biogas is emitted directly into atmosphere, the equivalent amount of power would have been supplied by CCPG, which is fed mainly by fossil fuel fired plants.

Applus+ Certification confirms that the Project has been approved by P. R. China government by checking the Project approval. By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in P. R. China.

3.3.5 Additionality

As per “Demonstration of additionality of small-scale project activities” version 13.1, additionality of the project shall be demonstrated by providing an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- (a) Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions;
- (b) Technological barrier: a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions;
- (c) Barrier due to prevailing practice: prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions;
- (d) Other barriers: without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

In line with the above guidance, the additionality is demonstrated using option (a) Investment barrier. The tool available in “Tool for the demonstration and assessment of additionality, version 07.0.0” is used to conduct the investment analysis and determine appropriate analytical methods.

The IRR calculation process was presented and confirmed as below:

The parameters used in the financial calculations have been validated based on a review of the sources presented in the PD, including: FSR⁴, Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects, etc. The FSR of the project was completed by a qualified third-party design institute, Henan Baichuan Changyin Environmental Protection Energy Co. LTD. Additionally, confirmation with the following documents and/or sources demonstrates that the parameters are plausible and acceptable under the project situation:

⁴ Only to the biogas capture and biogas generator units.

- Income Tax Law of the Peoples Republic of China for Enterprises issued by the Chairman of the People's Republic of China;
- Interim Regulation of VAT of the Peoples Republic of China issued by State Council etc.

Choice of approach

As the project generates financial and economic benefits other than VCS related income through the sales of fertilizers and other products and the alternative for the baseline scenario of the project does not involve an investment for the project participants, a benchmark analysis is justified for conducting the investment analysis of the project activity.

Benchmark Selection

The IRR benchmark (8%) used for the financial analysis in the FSR was obtained from the Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects. The benchmark IRR of capital Investment IRR (after tax) for power generation industry in China is 8%. As project activity involve in the power generation in China, then 8% as benchmark can be applied to this project.

Thus, the assessment team was able to confirm the suitability of this benchmark.

Crosscheck of input parameters

The assessment team has verified and crosschecked all the input values used for the IRR calculations in the PD submitted for registration. The input parameters used in the financial analysis are sourced from FSR.

For project, an approval letter of the FSR is issued by the government only after it passes the assessment of the sector experts designated by the government. It is in the assessment team's opinion that a FSR can be regarded as a reliable and trustworthy source of information coming from a recognized entity once it has been approved by the government. The FSR for the project was developed by a qualified designing institute Henan Baichuan Changyin Environmental Protection Energy Co. LTD. and approved by local government on 16/09/2021.

The assessment team compared the values stated in the PD with values determined in the FSR and was able to confirm that the input values are applied correctly in the PD.

Financial Parameters for project activity

Parameters	Unit	Value	Source
Total static investment	Million RMB	17.53	FSR
Installed capacity	MW	3	FSR

Net electricity supplied to the grid	MWh	11,847	FSR
Electricity Tariff (with VAT)	RMB/kWh	0.3779	FSR
VAT for electricity	%	17	FSR
Urban maintenance and construction tax	%	7	FSR
Education additional tax	%	3	FSR
Income tax	%	25	FSR
Lifetime	Year	10	FSR
Residual value rate of fixed assets	%	5	FSR
Annual operation & maintenance costs in a normal year	Million RMB	2.339	FSR
Expected VCUs price	RMB /tCO ₂ e	24	Assumption

The accuracy and suitability of the input values for the investment analysis were crosschecked as below:

Total static investment

The total static investment estimated in FSR is 17.53 million RMB, the same data is stated in the FSR approval issued by local government on 16/09/2021.

By checking audit of financial accounts of the project, the assessment team confirmed that the actual total static investment of the project activity was 16.887 million RMB, which is less than the value of static total investment in the FSR. But when apply this value in the IRR calculation, the IRR is still under the benchmark. Therefore, the assessment team was able to confirm that the total static investment in the FSR was acceptable.

O&M cost

The O&M cost estimated in FSR is 2.339 million RMB/yr. Considering the FSR was compiled by an accredited third party institution and approved by the government, the assessment team consider the value indicated in the FSR is accurate. Moreover, given the project has started operation more than 1 year, the assessment team checked O&M for year 2022 and confirmed

the actual O&M cost in 2022 is around 2.974 million RMB which is higher than the same in the FSR. Therefore, the assessment team was able to confirm that O&M cost in the FSR was reasonable and conservative.

Net electricity supplied to the grid

The net electricity supplied to the grid estimated in FSR is 11,847 MWh. Considering the FSR was compiled by an accredited third party institution and approved by the government, the assessment team consider the value indicated in the FSR is accurate. The plant load factor (PLF) of the project has been calculated under the EB 48 Annex 11. The guidelines for the reporting and validation of PLF, is based on FSR of this project and was determined by net electricity supplied to the grid, installed capacity of the project and 8,760 hour calculated as below:

$$PLF = 11,847 \text{ MWh} / 3 \text{ MW} / 8,760 \text{ h} = 45.08\%.$$

The net electricity supplied to the grid is sourced from FSR, and the installed capacity was verified by site visit and checking project approval. Therefore, the assessment team the PLF of project is reasonable and meet the requirement of guideline.

Thus, the assessment team was able to confirm that Net electricity supplied to the grid in the FSR was reasonable and accurate. Also this is verified by the actual net electricity supplied to the grid the project in 2022.

Electricity Tariff

The electricity tariff in FSR is 0.3779 RMB/kWh. Considering the FSR was compiled by an accredited third party institution and approved by the government, the assessment team consider the value indicated in the FSR is accurate. Moreover, by checking the Power Purchase Agreement (PPA) signed by project owner and power grid company, it is confirmed the actual electricity tariff of the project is 0.3779 RMB/kWh which is the same as FSR. Therefore, the assessment team was able to confirm that electricity tariff in the FSR was accurate.

Taxes

The taxes applied in the financial assessment are listed as:

Items	Value
VAT for electricity	17%
Urban construction tax and educational surtax	7%
Income tax	25%

VAT for electricity is determined as 17% in the FSR which is consistent with that in “Order of the State Council of the People's Republic of China (No. 691)” issued on 19/11/2017.

By cross checking with the national taxation regulations, sectoral regulations and other evidences, i.e. the “Interim Rules on Additional Tax for City Development” and the “Provisional Regulations of the People's Republic of China on Education tax”, the assessment team was able to confirm that the Urban construction tax and educational surtax in the financial assessment are all in line with national regulations.

An income tax of 25% is consistent with the Income Tax Law of the Peoples Republic of China or Enterprises issued by the Chairman of the People's Republic of China on 16/03/2007 and effective since 01/01/2008.

Depreciation period and residual rate: 10 years and 5%

As per the FSR, the residual rate is determined as 5% of the total static investment and the depreciation period is 10 years. As per Article 31 of the "Rules for the Provisional Regulations on Enterprise Income Tax of the People's Republic of China", the ratio of residual value shall be within 5% of the original value and shall be determined by the enterprise. Therefore, the assessment team was able to confirm that a rate of 5% is reasonable.

According to the Enforcement Regulation of Income Tax Law of the Peoples Republic of China for Enterprises issued by State Council on 06/12/2007 and effective since 01/01/2008, it stipulates that the depreciation period of production equipments should be at least 10 years. Thereby the 10 years of depreciation periods of the project activity is appropriate.

The depreciation rate of 9.5% is determined based on a depreciation period of 10 years and a residual rate of 5%.

Lifetime: 10 years

The lifetime of the project activity was estimated to be 10 years in the FSR. The lifetime of the project activity has been crosschecked with the technical specification of equipment, in which the lifetime of the equipment is 10 years.

Conclusion

Based on the information verified, the assessment team confirms that all the input values for investment analysis are in reasonable ranges. In addition, the assessment team was able to confirm that the input parameters used in the financial analysis are reasonable and adequately represent the economic situation of the project.

Calculation and conclusion

The IRR calculation was provided in the calculation spreadsheet. The project IRR (after tax) is demonstrated to be 3.40% without carbon revenues, which confirms that the project is

unattractive without the carbon revenues by comparing with benchmark value 8%. Therefore, the project IRR analysis is considered correct.

The spreadsheet of IRR calculation has been checked by the assessment team with the support of financial expertise. The calculation of financial figures is correctly done. And the calculation is in compliance with latest version of Investment Analysis;

- Technical lifetime takes 10 years which is in the range of 10-20 years;
- Project IRR (after tax) is used, and the Project IRR benchmark (after tax) is compared;
- The Recovery of fluid capital and recovery of fixed assets residual value at the end of the assessment period are considered as cash flow-in in the financial analysis;
- The depreciation cost is not included in the O&M cost for avoiding double counting;
- The financial cost (i.e. the loan payment) is not included in the O&M cost;
- All formulae used in the analysis in the spreadsheet are readable and all relevant cells are viewable and unprotected.

Therefore, the assessment team confirms that the assumptions in investment analysis are appropriate and the financial calculations are correct.

Sensitivity analysis

4 variables for, i.e. Total static investment, O&M cost, Net electricity supplied to the grid and Electricity Tariff which constitute more than 20% of either total project costs or total project revenues are considered in the sensitivity analysis. As verified, no other variables would constitute more than 20% of either total project costs or total project revenues. A departure variations of a range of +10% and -10% is applied which is consistent latest version of Investment Analysis.

As a result of sensitivity analysis, within the reasonable variation ($\pm 10\%$) of above variables, the IRR of project without VCS revenues is still lower than the benchmark.

Meanwhile, the assessment team assessed the possibility of the extreme points of the 4 variables to reach the benchmark 8%.

Total static investment: If the total static investment decreased by 24.96%, the IRR would reach 8%. However, it has been validated the actual total static investment of the project activity was 16.887 million RMB, which is less than the value of static total investment in the FSR. But when apply this value in the IRR calculation, the IRR is still under the benchmark. Therefore, it's impossible that the total static investment decreased by 24.96%.

O&M cost: Even O&M cost decreased to 0, the IRR of the project would still below benchmark. This is obviously not possible.

Net electricity supplied to the grid: If the Net electricity supplied to the grid increased by 16.11%, the IRR would reach 8%. Considering this value is determined by FSR compiled by third accredited institution and approved by government. Moreover, the net electricity supplied to the grid is calculated based on captured biogas, which is estimated conservatively. It is very unlikely for the project to become commercially attractive through an increase of the electricity generation.

Electricity Tariff: If the Electricity Tariff increased by 16.11%, the IRR would reach 8%. By checking the Power Purchase Agreement (PPA) signed by project owner and power grid company, it is confirmed the actual electricity tariff of the project is 0.3779 RMB/kWh which is the same as FSR. Therefore, it's unlikely that the Electricity Tariff increased by 16.11%.

It is assessment team's conclusion that sensitivity analysis used is appropriate and reasonable.

In conclusion, the additionality of the project has been justified, the project is additional.

3.3.6 Quantification of GHG Emission Reductions and Removals

Assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline emissions

According to methodology AMS-III.H version 19.0, baseline emissions for the systems affected by the project activity may consist of:

- (a) Emissions on account of electricity or fossil fuel used ($BE_{power,y}$);
- (b) Methane emissions from baseline wastewater treatment systems ($BE_{ww,treatment,y}$);
- (c) Methane emissions from baseline sludge treatment systems ($BE_{s,treatment,y}$);
- (d) Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/ lake/ sea ($BE_{ww,discharge,y}$);
- (e) Methane emissions from the decay of the final sludge generated by the baseline treatment systems ($BE_{s,final,y}$).

$$BE_y = BE_{power,y} + BE_{ww,treatment,y} + BE_{s,treatment,y} + BE_{ww,discharge,y} + BE_{s,final,y} \quad \text{Equation (1)}$$

Where:

BE_y Baseline emissions in year y (tCO₂e)

$BE_{power,y}$ Baseline emissions from electricity or fuel consumption in year y (tCO₂e)

$BE_{ww,treatment,y}$	Baseline emissions of the wastewater treatment systems affected by the project activity in year y (tCO ₂ e)
$BE_{s,treatment,y}$	Baseline emissions of the sludge treatment systems affected by the project activity in year y (tCO ₂ e)
$BE_{ww,discharge,y}$	Baseline methane emissions from degradable organic carbon in treated wastewater discharged into sea/river/lake in year y (tCO ₂ e).
$BE_{s,final,y}$	Baseline methane emissions from anaerobic decay of the final sludge produced in year y (tCO ₂ e). If the sludge is controlled combusted, disposed in a landfill with biogas recovery, or used for soil application in the baseline scenario, this term shall be neglected

By checking Project approval and on-site audit, it is confirmed that the project:

- The sludge is controlled combusted by sending to incinerator;
- In the baseline scenario, treated wastewater is used for factory reusing and is not discharged into sea/lake/river;
- There is no electricity consumption.

Therefore, equation (1) can be reduced to:

$$BE_y = BE_{ww,treatment,y} \quad \text{Equation (2)}$$

Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline plant as follows:

$$BE_{ww,treatment,y} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4} \quad \text{(Equation 3)}$$

Where:

$Q_{ww,i,y}$	Volume of wastewater treated in baseline wastewater treatment system i in year y (m ³). For ex ante estimation, forecasted wastewater generation volume or the designed capacity of the wastewater treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated wastewater
$COD_{inflow,i,y}$	Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y (t/m ³). Average value may be used through sampling with the confidence/precision level 90/10. Monitored values of

the COD inflow during crediting period will be used to calculate the baseline emissions ex post

$\eta_{\text{COD,BL},i}$	COD removal efficiency of the baseline treatment system i
$\text{MCF}_{\text{ww,treatment,BL},i}$	Methane correction factor for baseline wastewater treatment systems i . MCF value is identified as 0.8 corresponding to anaerobic deep lagoon (depth more than 2 meters) (IPCC and methodology)
i	Index for baseline wastewater treatment system
$B_{0,\text{ww}}$	Methane producing capacity of the wastewater (IPCC value of 0.25 kgCH ₄ /kg COD)
UF_{BL}	Model correction factor to account for model uncertainties (0.89)
GWP_{CH_4}	Global Warming Potential for methane

In determining baseline emissions using equation (1), historical records of at least one year prior to the project implementation shall be used. Considering the project started operation since 21/01/2022, one whole year historical record in 2021 before the implementation of project was used for determination of baseline emissions.

Calculation of $\text{BE}_{\text{ww,treatment},y}$

Applying the following values for ex-ante calculations:

Parameter	Value applied	Source
$Q_{\text{ww},i,y}$	313,608 m ³	Based on design data in the EIA
$\text{COD}_{\text{inflow},i,y}$	0.056805 t/m ³	Based on a 1-year historical record (operation log)
$\eta_{\text{COD,BL},i}$	76.33%	Based on a 1-year historical record (operation log)
$\text{MCF}_{\text{ww,treatment,BL},i}$	0.8	Corresponding to anaerobic deep lagoon (depth more than 2 meters)

$B_{0,ww}$	0.25 kgCH ₄ /kg COD	Default value as per the methodology
UF_{BL}	0.89	Default value as per the methodology
GWP_{CH_4}	28 tCO ₂ /tCH ₄	IPCC default value

Methane emissions from the baseline wastewater treatment system affected by the project $BE_{ww,treatment,y}$ are thus calculated as **67,771 tCO₂e/year**.

According to AMS-I.D version 18.0, baseline emissions from electricity displacement are calculated as follows:

$$BE_y = EG_{PJ,y} * EF_{grid,y}$$

Where:

BE_y Baseline emissions in year y (tCO₂)

$EG_{PJ,y}$ Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project in year y (MWh)

$EF_{grid,y}$ Combined margin CO₂ emission factor for grid connected power generation in year y (tCO₂/MWh)

The proposed project is the installation of a greenfield plant, so:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Where:

$EG_{PJ,facility,y}$ Quantity of net electricity generation supplied by the project plant to the grid in year y (MWh)

The grid emission factor is calculated as the weighted average of the operating margin (0.5) & build margin (0.5) values. The value of combined margin is sourced from 2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA. China DNA calculates the data based on Tool to Calculate the Emission Factor for an Electricity System", Version 07.0. No further assessment is required for grid emission calculation as the ex-ante value is sourced directly from the Chinese DNA.

$$\text{Emission factor (EF}_y\text{): } EF_y = EF_{grid,CM,y} = 0.8587 \times 0.5 + 0.2854 \times 0.5 = 0.57205 \text{ tCO}_2/\text{MWh}.$$

Then the baseline emissions from electricity displacement of the project is determined to be **6,777 tCO₂e/year**.

Project emissions

According to the methodology, project emissions may consist of:

- (a) CO₂ emissions from electricity and fuel used by the project facilities ($PE_{power,y}$),
- (b) Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww,treatment,y}$),
- (c) Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s,treatment,y}$),
- (d) Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww,discharge,y}$),
- (e) Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s,final,y}$),
- (f) Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive,y}$),
- (g) Methane emissions due to incomplete flaring ($PE_{flaring,y}$),
- (h) Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass,y}$)

By checking FSR, project approval and on-site audit, it is confirmed that the project activity does not involve project emissions as below:

There is no flare used to destroy the leachate biogas, therefore, $PE_{flaring,y}$ is equal to 0.

The sludge is controlled combusted by sent to incinerator. Project activity does not involve sludge treatment, therefore, $PE_{s,treatment,y}$ and $PE_{s,final,y}$ is equal to 0.

These emissions shall be calculated as per paragraph 26 of the methodology, If recovered biogas in the baseline is used to power auxiliary equipment it should be taken into account accordingly, using zero as its emission factor. The electricity used by project activity will be firstly supplied by the gas engines which use biogas as fuel, to import electricity from the grid is the second option when the gas engines does not operate. The project deliver the electricity to the project activity before delivery to the grid. Then the chance that the project activity will use electricity from grid is very low. Therefore, $PE_{power,y}$ can be considered as zero.

Only when the biogas plant need to import the electricity from grid the $PE_{power,y}$ will be taken into account.

The water treatment system remains unchanged, and only methane capture system was installed. Therefore, $PE_{ww,treatment,y}$, $PE_{biomass,y}$ and $PE_{ww,discharge,y}$ is equal to 0.

Thus, project emissions consist of:

$$PE_y = PE_{fugitive,y} \quad \text{Equation (4)}$$

Where:

PE_y Project activity emissions in the year y (tCO₂e)

$PE_{fugitive,y}$ Methane emissions from biogas release in capture systems in year y (tCO₂e)

Equations used for the calculation of $PE_{power,y}$

Project emissions from energy consumption are composed of 1) wastewater treatment system consumption and 2) system for biogas recovery and flaring/gainful use.

Electricity consumed for wastewater treatment system will not be affected by the proposed project activity, it will be the same in the baseline and in the project scenario, so emissions are not accounted for. This is conservative since in the baseline, electricity is sourced from the grid whereas in the project situation, electricity will be sourced from on-site generation from biogas.

It is expected that electricity consumed for the biogas recovery system and flaring/gainful use will be sourced from the onsite generation, therefore, $PE_{power,y}$ is expected to be zero.

In case it is consumed from another source than biogas recovered on site (grid or fossil-fuel), $PE_{power,y}$ will be calculated as per the Tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” as follows:

$$PE_{power,y} = PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y}) \quad \text{Equation (5)}$$

Where:

$PE_{EC,y}$ Project emissions from electricity consumption in year y (tCO₂/yr)

$EC_{PJ,j,y}$ Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)

$EF_{EF,j,y}$ Emission factor for electricity generation for source j in year y (tCO₂/MWh)

$TDL_{j,y}$ Average technical transmission and distribution losses for providing electricity to source j in year y

Calculation of $PE_{power,y}$

It is expected that electricity consumed for the operation of the project activity including biogas recovery will be sourced from on-site power generation from biogas. Therefore, for ex-ante calculations, **$PE_{power,y}$ is assumed to be zero.**

In case the electricity is consumed from another source (grid or fossil-fuel), $PE_{power,y}$ will be calculated as per the Tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” as follows:

$$PE_{power,y} = PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y})$$

Where:

$EF_{EF,j,y} = EF_{grid,CM,y} = 0.57205 \text{ tCO}_2\text{e/MWh}$, calculated by official sources according to the Tool to calculate the emission factor for an electricity system.

$TDL_{j,y} = 20\%$, default value proposed by Tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”

$EC_{PJ,j,y}$ will be monitored ex-post.

Equations used for the calculation of $PE_{fugitive,y}$

Project activity emissions from methane release in capture systems are determined as follows:

Based on the methane emission potential of wastewater:

$$PE_{fugitive,y} = PE_{fugitive,ww,y} \quad \text{Equation (6)}$$

Where:

$PE_{fugitive,ww,y}$ Fugitive emissions through capture inefficiencies in the anaerobic wastewater treatment systems in the year y (tCO_2e)

$$PE_{fugitive,ww,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH_4} \quad \text{Equation (7)}$$

Where:

CFE_{ww} Capture efficiency of the biogas recovery equipment in the wastewater treatment systems (a default value of 0.9 shall be used)

$MEP_{ww,treatment,y}$ Methane emission potential of wastewater treatment systems equipped with biogas recovery system in year y (tCH_4)

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{0,ww} \times UF_{PJ} \times \sum_k COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k} \quad \text{Equation (8)}$$

Where:

$COD_{removed,PJ,k,y}$ Chemical oxygen demand removed by the treatment system k of the project activity equipped with biogas recovery in the year y (t/m^3)

$MCF_{ww,treatment,PJ,k}$ Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment (MCF value 0.8)

UF_{PJ} Model correction factor to account for model uncertainties (1.12)

Calculation of $PE_{fugitive,y}$

$$PE_{fugitive,ww,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH_4}$$

Applying the following values:

CFE_{ww} 0.9 (default value)

$MEP_{ww,treatment,y}$ Calculated as 2,942 tCH₄ (see details below)

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} \times UF_{PJ} \times \sum_k COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k}$$

Where:

Parameter	Value applied	Source
$Q_{ww,y}$	313,608 m ³	Based on design data in the EIA
$COD_{removed,PJ,k,y}$	0.041880 t/m ³	Based on a 1-year historical record (operation log)
$MCF_{ww,treatment,PJ,i}$	0.8	Corresponding to anaerobic digester (depth more than 2 meters)
$B_{o,ww}$	0.25 kgCH ₄ /kg COD	Default value as per the methodology
UF_{PJ}	1.12	Default value as per the methodology

$COD_{removed,PJ,k,y}$ is the difference between $COD_{inflow,PJ,y}$ and $COD_{outflow,PJ,y}$ of the treatment system installed in the Project.

Inflow and outflow of COD are regularly measured and monitored by project proponent. The values applied for ex-ante calculations of project emissions are the average values of COD_{inflow} and $COD_{outflow}$ measured based on a 1-year historical record in 2021.

Parameter	Result
$COD_{inflow,PJ,y}$	0.056805 t/m ³
$COD_{outflow,PJ,y}$	0.014924 t/m ³
$COD_{removed,PJ,k,y}$	0.041880 t/m ³

Therefore, $PE_{fugitive,ww,y} = (1 - 0.9) \times 2,942 \times 28 = 8,238 \text{ tCO}_2$

Ex-ante annual estimates of project emissions are calculated as $PE_y = 8,238 \text{ tCO}_2$.

Leakage

The project does not need to consider leakage. The validation team deems this consideration is correct and in line with methodology AMS-III.H, version 19.0.

Emission Reductions

According to AMS-III.H, Version 19.0, emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emissions in year y (tCO₂e/yr)

PE_y = Project emissions in year y (tCO₂e/yr)

Hence for the project activity, the estimated amount of GHG emission reductions (ER_y) is 663,100 tCO₂e during the crediting period from 21/01/2022 to 20/01/2032, resulting in estimated average annual emission reductions of 66,310 tCO₂e.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
21/01/2022-31/12/2022	70,463	7,786	0	62,677
2023	74,548	8,238	0	66,310
2024	74,548	8,238	0	66,310
2025	74,548	8,238	0	66,310
2026	74,548	8,238	0	66,310
2027	74,548	8,238	0	66,310
2028	74,548	8,238	0	66,310
2029	74,548	8,238	0	66,310

2030	74,548	8,238	0	66,310
2031	74,548	8,238	0	66,310
01/01/2032- 20/01/2032	4,084	451	0	3,633
Total	745,479	82,379	0	663,100

3.3.7 Methodology Deviations

No methodology deviation is applied in the project.

3.3.8 Monitoring Plan

Assessment team checked the monitoring practice through on-site audit and also checked the guideline of respective State electricity regulatory commission. The detail analysis is as below:

Parameters determined ex-ante

Data / Parameter:	$\eta_{\text{COD,BL},i}$
Data unit:	%
Description:	COD removal efficiency of the baseline treatment system
Source of data used:	Operation log of wastetreatment
Value applied:	76.33

Data / Parameter:	GWP_{CH_4}
Data unit:	$\text{tCO}_2\text{e/t CH}_4$
Description:	Global warming potential of methane
Source of data used:	IPCC

Value applied:	28
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Data / Parameter:	$MCF_{www,treatment,BL,i}$
Data unit:	-
Description:	Methane Correction Factor for baseline wastewater treatment system i
Source of data used:	AMS-III.H
Value applied:	Default value as 0.8 for anaerobic deep lagoon (depth more than 2 meters)

Data / Parameter:	$MCF_{www,treatment,PJ,k}$
Data unit:	-
Description:	Methane Correction Factor for project wastewater treatment system k
Source of data used:	AMS-III.H
Value applied:	Default value as 0.8 for anaerobic digester (depth more than 2 meters)

Data / Parameter:	$B_{o,ww}$
Data unit:	kg CH ₄ /kg COD
Description:	Methane producing capacity of the wastewater

Source of data used:	IPCC
Value applied:	Default value as 0.25

Data / Parameter:	UF _{BL}
Data unit:	-
Description:	Model correction factor to account for model uncertainties
Source of data used:	AMS-III.H
Value applied:	Default value as 0.89

Data / Parameter:	UF _{PJ}
Data unit:	-
Description:	Model correction factor to account for model uncertainties
Source of data used:	AMS-III.H
Value applied:	Default value as 1.12

Data / Parameter:	CFE _{ww}
Data unit:	-
Description:	Capture efficiency of the biogas recovery equipment in the wastewater treatment system
Source of data used:	AMS-III.H

Value applied:	Default value as 0.9
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Data / Parameter:	D _{CH4}
Data unit:	tCH ₄ /m ³
Description:	Methane density
Source of data used:	-
Value applied:	0.0007168

Data / Parameter:	FE
Data unit:	%
Description:	Flare efficiency in year y
Source of data used:	AMS-III.H
Value applied:	100

Data / Parameter:	EF _{grid,OM,y}
Data unit:	tCO ₂ /MWh
Description:	Operation margin emission factor of CCPG
Source of data used:	2019 Baseline Emission Factors for Regional Power Grids in China
Value applied:	0.8587

Data / Parameter:	$EF_{grid,BM,y}$
Data unit:	tCO ₂ /MWh
Description:	Build margin emission factor of CCPG
Source of data used:	2019 Baseline Emission Factors for Regional Power Grids in China
Value applied:	0.2854

Data / Parameter:	$TDL_{j,y}$
Data unit:	%
Description:	Average technical transmission and distribution losses for providing electricity to source j in year y
Source of data used:	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)
Value applied:	Use as default values of 20% for project consumption sources. i.e.: $TDL_{j,y} = 20\%$

Parameters determined ex-post

- Wastewater Treatment: $Q_{ww,i,y}$, $COD_{ww,untreated,y}$ and $COD_{ww,treated,y}$

Parameters title	Descriptions
$Q_{ww,i,y}$	The flow of wastewater
$COD_{ww,untreated,y}$	The chemical oxygen demand of the wastewater before the treatment system affected by the project activity

$COD_{ww,treated,y}$	The chemical oxygen demand of the wastewater after the treatment system affected by the project activity
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$Q_{ww,i,y}$ will be continuous measured by 2 flow meters installed before entering the treatment system. The flow meters will be calibrated in accordance with national standards or other relevant requirements. The accuracy of flow meters will be satisfied with relevant standards or requirements.

$COD_{ww,untreated,y}$ and $COD_{ww,treated,y}$ would be measured through spectrophotometric method according to national or international standards on a daily basis.

- Biogas Generation: $BG_{burnt,y}$, WCH_4,y , T and P

Parameters title	Descriptions
$BG_{burnt,y}$	Biogas volume in year y
WCH_4,y	Methane content in biogas in year y
T	Temperature of the biogas
P	Pressure of the biogas

$BG_{burnt,y}$ will be continuous measured by flowmeter(s). The flow meter(s) will be calibrated in accordance with national standards or other relevant requirements. The accuracy of flow meter(s) will be satisfied with relevant standards or requirements.

WCH_4,y will be measured with a continuous gas analyzer(s). The gas analyzer(s) will be calibrated periodically in accordance with manufacturer's specifications or relevant requirements. The accuracy of gas analyzer(s) will be satisfied with manufacturer's specifications or relevant requirements.

T will be measured by temperature sensor(s). The temperature sensor(s) will be calibrated periodically in accordance with manufacturer's specifications or relevant requirements. The accuracy of temperature sensor(s) will be satisfied with manufacturer's specifications or relevant requirements. But if the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalized flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas.

P will be measured by pressure sensor(s). The pressure sensor(s) will be calibrated periodically in accordance with manufacturer's specifications or relevant requirements. The accuracy of

pressure sensor(s) will be satisfied with manufacturer's specifications or relevant requirements. But if the biogas flow meter employed measures flow, pressure and temperature and displays or outputs the normalized flow of biogas, then there is no need for separate monitoring of pressure and temperature of the biogas.

- Electricity Generation: $EC_{PJ,y}$, $EG_{PJ,facility,y}$

Parameters title	Descriptions
$EC_{PJ,y}$	Quantity of electricity consumed by the project in year y
$EG_{PJ,facility,y}$	Quantity of net electricity generation supplied by the project to the grid in year y

$EC_{PJ,y}$ and $EG_{PJ,facility,y}$ will be continuous measured by electricity meters and monthly recorded. The electricity meters will be calibrated in accordance with national standards or other relevant requirements. The accuracy of electricity meters will be satisfied with relevant standards or requirements.

Based on the on-site audit and interviewed with the Manager of Zhongmu Branch Company of Henan BCCY Environmental Energy Co., Ltd., the validation team confirmed that monitoring parameter has been correctly described in the PD and in compliance with the methodology AMS-III.H Version 19.0, AMS-I.D Version 18.0 and the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and VCS via VCS standard version 4.4.

3.4 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VALIDATION CONCLUSION

Applus+ Certification has been engaged by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project.

The management of the project proponent/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project's Monitoring Plan in the VCS PD and the approved methodology AMS-III.H version 19.0 and AMS-I.D version 18.0.

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board and VERRA registry. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

The projects description compliance with, the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS guideline and standard version 4.4.

The project's baseline and additionality is assessed against AMS-III.H version 19.0 and AMS-I.D version 18.0 for small scale project.

The project's monitoring plan is assessed against AMS-III.H version 19.0 and AMS-I.D version 18.0 for small scale project.

A risk based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

APPENDIX I: <REFERENCE LIST>

1. VCS project design version 01 dated 26/09/2022, version 04 dated 25/06/2023
2. Estimated ER calculation spreadsheet
3. VCS standard version 4.4
4. VCS guideline version 4.3
5. CDM Validation and Verification Standard for project activities version 03.0
6. Approved methodology AMS-III.H: Methane recovery in wastewater treatment, version 19.0
AMS-I.D: Grid connected renewable electricity generation, version 18.0
7. Tool to calculate the emission factor for an electricity system, version 07.0
8. Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
9. Demonstration of additionality of small-scale project activities, version 13.1
10. Tool for the demonstration and assessment of additionality, version 07.0.0
11. Business License
12. Feasibility Study Report
13. Project approval
14. Environmental Impact Assessment Report
15. Environmental Impact Assessment (EIA) approval
16. Cooperation agreement

17. Operation Log of project activity
18. Nameplate of the equipment
19. Operation log of Dongxing Waste Incineration Plant in 2021
20. 2019 Baseline Emission Factors for Regional Power Grids in China dated 29/12/2020 published by China DNA
21. Questionnaires for stakeholder meeting
22. Interim Rules on Economic Assessment of Electrical Engineering Retrofit Projects
23. Income Tax Law of the Peoples Republic of China for Enterprises issued by the Chairman of the People's Republic of China
24. Interim Regulation of VAT of the Peoples Republic of China issued by State Council
25. Audit of financial accounts of the project
26. O&M for year 2022
27. Power Purchase Agreement (PPA)
28. Order of the State Council of the People's Republic of China (No. 691)
29. Interim Rules on Additional Tax for City Development
30. Provisional Regulations of the People's Republic of China on Education tax
31. Rules for the Provisional Regulations on Enterprise Income Tax of the People's Republic of China
32. Enforcement Regulation of Income Tax Law of the Peoples Republic of China for Enterprises
33. Technical specification of equipment

APPENDIX II: <CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)>

CAR ID	01	Section no.	4	Date: 30/12/2022
Description of CAR				
The emission reduction calculation of project is not conservative.				
Project proponent response				Date: 14/02/2023
The calculation of emission reduction has been conducted in most conservative manner.				
Documentation provided by project proponent				
Updated PD and ER spreadsheet				
VVB assessment				Date: 15/02/2023
By checking updated PD and ER spreadsheet, it is confirmed that the calculation of emission reduction has been conducted in most conservative manner.				
CAR01 is closed out.				

CAR ID	02	Section no.	5.2	Date: 30/12/2022
Description of CAR				
The monitoring plan is incomplete comparing with requirement of methodology.				
Project proponent response				Date: 14/02/2023

Missing parameters have been included in the updated PD.	
Documentation provided by project proponent	
Updated PD	
VVB assessment	Date: 15/02/2023
By checking updated PD, it is confirmed that missing parameters have been included in the updated PD and in line with the requirement of methodology. CAR02 is closed out.	